

A Mathematical Derivations and Properties

A.1 Weighted Excess Branching

Let i be a node with effective nonnegative incident edge weights $\{p_{ij} : j \in N(i)\}$, and weighted strength $s_i = \sum_{j \in N(i)} p_{ij}$. We seek a weighted analogue of the classical excess degree $k_i - 1$, interpreted as the residual outward connectivity available after arriving at i .

Assume that arrival at i occurs through edge (j, i) with probability proportional to its interaction strength,

$$\Pr(J = j \mid i) = \frac{p_{ij}}{s_i}, \quad s_i > 0.$$

Conditional on arrival through j , the residual weighted strength available without immediately returning along the same edge is $s_i - p_{ij}$. Therefore, the expected residual strength is

$$\begin{aligned} b_i &= \sum_{j \in N(i)} \frac{p_{ij}}{s_i} (s_i - p_{ij}) \\ &= s_i - \frac{1}{s_i} \sum_{j \in N(i)} p_{ij}^2. \end{aligned}$$

Hence,

$$b_i = s_i - \frac{\sum_{j \in N(i)} p_{ij}^2}{s_i}$$

for $s_i > 0$, and we set $b_i = 0$ when $s_i = 0$.

Equivalently,

$$b_i = s_i \left(1 - \sum_{j \in N(i)} \left(\frac{p_{ij}}{s_i} \right)^2 \right).$$

This expression shows that b_i depends not only on total strength but also on how that strength is distributed across incident edges. Concentration of most weight on a single edge reduces residual branching because arrival is then likely to consume a large fraction of the node's available interaction strength. The construction satisfies $0 \leq b_i \leq s_i$, gives $b_i = 0$ for a weighted leaf, and reduces to $k_i - 1$ under unit weights. For fixed strength, b_i decreases as edge weight becomes more concentrated, distinguishing it from the naive substitution $s_i - 1$. The weighted branching term satisfies several useful properties.

Nonnegativity. Since $0 \leq p_{ij} \leq s_i$,

$$p_{ij}^2 \leq s_i p_{ij}.$$

Summing over j gives $\sum_j p_{ij}^2 \leq s_i^2$, and therefore

$$b_i = s_i - \frac{\sum_j p_{ij}^2}{s_i} \geq 0.$$

Upper bound. Because the second term is nonnegative,

$$0 \leq b_i \leq s_i.$$

Leaf consistency. If i has exactly one incident edge with weight $p > 0$, then $s_i = p$ and

$$b_i = p - \frac{p^2}{p} = 0.$$

Thus a weighted leaf has no residual outward branching, matching the unweighted interpretation.

Unit-weight reduction. If $p_{ij} = 1$ for every $j \in N(i)$, then $s_i = k_i$ and $\sum_j p_{ij}^2 = k_i$. Therefore,

$$b_i = k_i - \frac{k_i}{k_i} = k_i - 1.$$

Hence the classical excess-degree factor is recovered exactly.

Effect of weight concentration. For fixed s_i , the quantity b_i decreases as $\sum_j p_{ij}^2$ increases. Thus two nodes with identical weighted strength may have different residual branching: interaction strength distributed across many edges produces larger b_i than the same total strength concentrated on a small number of edges.

This property is why the naive expression $s_i - 1$ is not generally appropriate. Subtracting one unit of strength assumes that arrival always consumes exactly one unit, regardless of the edge used. In a heterogeneous weighted graph, the expected consumed strength is instead $\sum_j p_{ij}^2 / s_i$.

A.2 Derivation of Weighted Collective Influence

Classical collective influence combines the excess degree of a focal node with the excess degree of nodes reached at a prescribed radius. In the weighted setting, two modifications are required. First, $k_i - 1$ is replaced by the weighted excess branching term b_i . Second, propagation along a sequence of edges should depend on the strengths of those edges.

Let

$$P = (v_0, v_1, \dots, v_\ell), \quad v_0 = i,$$

be a length- ℓ non-backtracking walk, meaning $v_{r+1} \neq v_{r-1}$ for every admissible r . Its weighted transmission factor is

$$w(P) = \prod_{r=0}^{\ell-1} p_{v_r v_{r+1}}.$$

The contribution of P to the outward influence of i is taken as the product of three components: the residual branching at the source b_i , the interaction strength accumulated along the walk $w(P)$, and the residual branching at its endpoint b_{v_ℓ} . Summing over all length- ℓ non-backtracking walks gives

$$WCI_\ell(i) = b_i \sum_{P \in \mathcal{W}_\ell^{NB}(i)} \left(\prod_{e \in P} p_e \right) b_{\text{end}(P)}.$$

The construction discounts weak propagation routes multiplicatively. A walk containing a weak edge contributes less even if its endpoint has large branching potential.

For $\ell = 1$, the expression simplifies to

$$WCI_1(i) = b_i \sum_{j \in N(i)} p_{ij} b_j.$$

For $\ell = 2$,

$$WCI_2(i) = b_i \sum_{j \in N(i)} p_{ij} \sum_{\substack{x \in N(j) \\ x \neq i}} p_{jx} b_x.$$

The exclusion $x \neq i$ is the non-backtracking condition and prevents an immediate reversal along (i, j) .

A.3 Relationship to Classical Collective Influence

The classical collective-influence score at radius ℓ is commonly written

$$CI_\ell(i) = (k_i - 1) \sum_{j: d(i, j) = \ell} (k_j - 1),$$

where the summation is over the shortest-distance sphere of radius ℓ around i .

For unit weights and $\ell = 1$, WCI reduces exactly to classical CI_1 . Indeed, $p_{ij} = 1$ implies $b_i = k_i - 1$, so

$$\begin{aligned} WCI_1(i) &= (k_i - 1) \sum_{j \in N(i)} (k_j - 1) \\ &= CI_1(i). \end{aligned}$$

For $\ell > 1$, the two quantities need not coincide on graphs containing cycles. WCI sums over non-backtracking walks, whereas classical CI_ℓ sums each node on the shortest-distance sphere once. A node may therefore contribute through more than one WCI walk when multiple non-backtracking routes reach it.

On tree-like neighborhoods, where each radius- ℓ node is reached by a unique simple path from i , the distinction disappears under unit weights, and the WCI expression coincides with the corresponding sphere-based collective-influence calculation. On general cyclic graphs, however,

$$WCI_\ell \neq CI_\ell \quad \text{in general for } \ell > 1.$$

This distinction is deliberate. WCI is formulated as a weighted non-backtracking propagation measure, rather than as a weighted substitution inside the shortest-distance definition of classical CI.

A.4 Non-Backtracking Message-Passing Formulation

Direct enumeration of all non-backtracking walks is unnecessary. The WCI score can be computed using directed-edge messages.

For each directed edge $i \rightarrow j$, define

$$M_{i \rightarrow j}^{(0)} = b_j.$$

For $r \geq 1$, $M_{i \rightarrow j}^{(r)}$ represents the total weighted endpoint branching contribution of length- r non-backtracking continuations starting from j , conditional on having arrived from i . It satisfies

$$M_{i \rightarrow j}^{(r)} = \sum_{\substack{x \in N(j) \\ x \neq i}} p_{jx} M_{j \rightarrow x}^{(r-1)}.$$

To avoid explicitly filtering $x = i$, define

$$A_j^{(r)} = \sum_{x \in N(j)} p_{jx} M_{j \rightarrow x}^{(r-1)}.$$

Then

$$\boxed{M_{i \rightarrow j}^{(r)} = A_j^{(r)} - p_{ji} M_{j \rightarrow i}^{(r-1)}}.$$

Finally,

$$\boxed{WCI_\ell(i) = b_i \sum_{j \in N(i)} p_{ij} M_{i \rightarrow j}^{(\ell-1)}}.$$

To verify the recurrence, consider the base case. For $\ell = 1$,

$$WCI_1(i) = b_i \sum_j p_{ij} M_{i \rightarrow j}^{(0)} = b_i \sum_j p_{ij} b_j,$$

which is the definition above.

Assume $M_{i \rightarrow j}^{(r-1)}$ correctly aggregates all length- $(r-1)$ non-backtracking continuations. Extending each continuation by one edge from j requires summing over all neighbors x except the node i from which the walk arrived. The recurrence therefore enumerates exactly the admissible non-backtracking extensions, and induction establishes equivalence to the walk-based definition for all r . Each propagation depth processes every directed edge a constant number of times. Thus WCI computation requires $O(\ell m)$ time and $O(m + n)$ memory.

B Theoretical Properties

This section establishes the basic properties of the W-CIFSA objective and greedy extraction procedure. These results concern boundedness, feasibility, local improvement, termination, and computational cost. They do not imply submodularity, global optimality, or a constant-factor approximation guarantee.

B.1 Boundedness of the Focal-Group Objective

Recall that

$$F_{\lambda, \kappa}(S) = \frac{(1 - \lambda) \sum_{i \in S} q_i + \frac{2\lambda}{s_{\max}} A(S)}{|S| + \kappa},$$

where $q_i \in [0, 1]$, $\lambda \in [0, 1]$, $\kappa \geq 0$, and $A(S)$ is the total weight of edges internal to S .

Proposition 1. For every nonempty $S \subseteq V$,

$$0 \leq F_{\lambda, \kappa}(S) \leq 1.$$

Proof. Nonnegativity follows directly from $q_i \geq 0$, $p_{ij} \geq 0$, and $\lambda, \kappa \geq 0$. For the upper bound, $\sum_{i \in S} q_i \leq |S|$. In addition,

$$2A(S) = \sum_{i \in S} \sum_{\substack{j \in S \\ (i, j) \in E}} p_{ij} \leq \sum_{i \in S} s_i \leq |S| s_{\max}.$$

Hence $2A(S)/s_{\max} \leq |S|$, and the numerator satisfies

$$(1 - \lambda) \sum_{i \in S} q_i + \frac{2\lambda}{s_{\max}} A(S) \leq (1 - \lambda)|S| + \lambda|S| = |S|.$$

Since $|S| + \kappa \geq |S|$, the result follows. $\square$

B.2 Connectivity Invariant

W-CIFSA restricts candidate additions to the frontier of the current group. For a connected set S , define its frontier as the unassigned nodes outside S that are adjacent to at least one node in S .

Proposition 2. If the initial group is connected and every accepted node is selected from its current frontier, then every intermediate and terminal W-CIFSA group is connected.

Proof. The initial set is connected by construction. Suppose S is connected and x is selected from its frontier. Then x has an edge to at least one node in S . Therefore $S \cup \{x\}$ is connected. Repeated application establishes the result by induction. $\square$

Thus connectivity is maintained as a hard feasibility condition throughout extraction; it is not inferred from the value of the objective.

B.3 Monotonicity of Accepted Expansions

For a valid frontier candidate x , define

$$K_S(x) = (1 - \lambda)q_x + \frac{2\lambda}{s_{\max}}a_x(S),$$

where $a_x(S) = \sum_{i \in S} p_{ix}$. The marginal objective change is

$$\Delta_{\lambda, \kappa}(x \mid S) = \frac{K_S(x) - F_{\lambda, \kappa}(S)}{|S| + \kappa + 1}.$$

Proposition 3. Under the zero-tolerance acceptance rule $K_S(x) > F_{\lambda, \kappa}(S)$, every accepted expansion strictly increases the objective.

Proof. Because the denominator $|S| + \kappa + 1$ is positive,

$$K_S(x) > F_{\lambda, \kappa}(S) \iff \Delta_{\lambda, \kappa}(x \mid S) > 0.$$

Therefore,

$$F_{\lambda, \kappa}(S \cup \{x\}) > F_{\lambda, \kappa}(S).$$

$\square$

Consequently, the objective values along an accepted expansion trajectory form a strictly increasing sequence.

B.4 Finite Termination

Proposition 4. W-CIFSA terminates after finitely many accepted expansions.

Proof. The graph contains a finite number $n = |V|$ of nodes. Each accepted step adds one previously unassigned node to the current group, and nodes are never removed during greedy expansion. Therefore a single group can contain at most n nodes and can undergo only finitely many augmentations.

Under disjoint extraction, completed-group nodes are removed from consideration for subsequent groups. Hence the total number of accepted node assignments across all groups is also bounded by n . If no valid improving frontier node exists, the current expansion terminates immediately. Thus the full procedure terminates in finite time. $\square$

B.5 One-Step Augmentation Local Optimality

Let ∂S denote the valid frontier of S under the current disjointness and feasibility constraints.

Proposition 5. Let S be a group at which W-CIFSA terminates under zero acceptance tolerance. Then

$$F_{\lambda,\kappa}(S \cup \{x\}) \leq F_{\lambda,\kappa}(S)$$

for every $x \in \partial S$.

Proof. W-CIFSA terminates only when its best valid frontier candidate x^* fails the acceptance condition. Since x^* maximizes $K_S(x)$ over $x \in \partial S$,

$$K_S(x) \leq K_S(x^*) \leq F_{\lambda,\kappa}(S)$$

for every valid frontier candidate x . The marginal-gain expression therefore gives

$$\Delta_{\lambda,\kappa}(x \mid S) \leq 0$$

for every $x \in \partial S$, proving the result. $\square$

This is a one-step augmentation property only. It does not imply optimality under deletion, node swaps, multi-node additions, or arbitrary connected-set replacement, and it does not imply global optimality. The exact-optimality experiments in Appendix ?? empirically quantify the resulting global objective gap.

B.6 Locality of Candidate-Utility Updates

The candidate-utility structure permits local rather than global score updates during expansion.

Proposition 6. If node y is added to S , then for any existing candidate $x \notin S \cup \{y\}$,

$$K_{S \cup \{y\}}(x) = K_S(x) + \frac{2\lambda}{s_{\max}} p_{xy}.$$

Proof. The influence component q_x is independent of the current group. The attachment term changes as

$$a_x(S \cup \{y\}) = a_x(S) + p_{xy}.$$

Substituting this expression into the definition of $K_S(x)$ gives the result. $\square$

For nodes not adjacent to y , $p_{xy} = 0$, so their candidate utilities are unchanged. Hence an accepted augmentation requires score updates only for neighbors of the newly added node. Newly exposed frontier nodes can likewise be initialized from their attachment to the current group.

This locality supports an exact versioned lazy max-heap implementation: stale heap entries are discarded when popped, while only candidates whose attachment values change require new keys.

B.7 Computational Complexity

Let $n = |V|$ and $m = |E|$. As shown in Appendix A.4, WCI computation requires $O(\ell m)$ time and $O(m + n)$ memory.

During group extraction, adding a node changes the candidate utility only for its neighbors. With versioned lazy-heap updates, these edge-local changes give $O(m \log n)$ time over a complete disjoint extraction pass. The first calibration pass and final extraction pass differ only by a constant factor.

Therefore, a complete two-pass W-CIFSA run requires

$$O(\ell m + m \log n)$$

time and $O(m + n)$ memory. For fixed ℓ , the dominant asymptotic term is $O(m \log n)$.

C Support Regularization and Calibration

C.1 Motivation for Support Regularization

The unregularized focal-group objective is $F_{\lambda,0}(S) = U_\lambda(S)/|S|$. Because this is an average utility, a small set can achieve a high score even when a slightly larger connected group has comparable total utility. In preliminary experiments, this behavior frequently produced two-node terminal groups, especially when the initial pair already contained high-influence nodes or a strong internal edge.

To reduce this small-set bias without introducing a fixed minimum group size, W-CIFSA uses the regularized objective

$$F_{\lambda,\kappa}(S) = \frac{U_\lambda(S)}{|S| + \kappa}, \quad \kappa \geq 0.$$

The regularization acts as a support discount: for fixed $U_\lambda(S)$, larger κ reduces the advantage of very small groups. Equivalently,

$$F_{\lambda,\kappa}(S) = F_{\lambda,0}(S) \frac{|S|}{|S| + \kappa}.$$

Thus $\kappa = 0$ recovers the original average objective, while positive κ progressively discounts small sets. Importantly, κ is not selected using diffusion outcomes; it is calibrated from the stopping behavior of the group objective itself.

C.2 Derivation of the Required Support Parameter

Consider a current group S with $|S| = s$ under the unregularized objective $\kappa = 0$. Let x^* be its best valid frontier candidate and define

$$r_{\text{stop}}(S) = \frac{K_S(x^*)}{F_{\lambda,0}(S)}.$$

At a terminal state, $r_{\text{stop}} \leq 1$. We ask how much support regularization would be required to make x^* exactly marginal.

With regularization κ , the candidate is marginal when $K_S(x^*) = F_{\lambda,\kappa}(S)$. Since

$$F_{\lambda,\kappa}(S) = F_{\lambda,0}(S) \frac{s}{s + \kappa},$$

we obtain

$$r_{\text{stop}} = \frac{s}{s + \kappa}.$$

Solving for κ gives

$$\kappa_{\text{req}}(S) = s \left(\frac{1}{r_{\text{stop}}(S)} - 1 \right).$$

The calibration is based specifically on two-node terminal groups, so $s = 2$, yielding

$$\kappa_{\text{req}}(S) = 2 \left(\frac{1}{r_{\text{stop}}(S)} - 1 \right).$$

A state with r_{stop} close to one requires only a small amount of regularization to reopen, whereas a state whose best candidate is far below the current objective requires substantially larger κ .

C.3 Pair-Terminal State Identification

Let $\mathcal{T}_2$ denote the set of groups that satisfy all of the following:

1. the group has exactly two nodes;
2. it terminates during the unregularized $\kappa = 0$ extraction;
3. at least one valid frontier candidate exists;
4. termination occurs because the best candidate fails the marginal improvement test.

Pairs that terminate because their frontier is empty are excluded, since no choice of κ can create a missing structural continuation. Likewise, termination caused by disjointness or other feasibility constraints is not treated as evidence of small-set bias.

For each $S \in \mathcal{T}_2$, we compute $r_{\text{stop}}(S)$. States with $r_{\text{stop}} = 0$ are retained for diagnostics but excluded from the finite κ_{req} distribution because their required support is unbounded. If no valid positive- r_{stop} pair-terminal states are observed, W-CIFSA sets $\kappa^* = 0$.

This distinction is important: the calibration responds only to observed marginal stopping behavior, not to all groups of size two.

C.4 Quantile-Based Calibration Rule

A single κ must be used for the final extraction pass. Rather than using the maximum required support, which can be dominated by extreme terminal states, W-CIFSA uses a robust upper quantile:

$$\kappa^* = Q_{0.75} \{ \kappa_{\text{req}}(S) : S \in \mathcal{T}_2, r_{\text{stop}}(S) > 0 \}.$$

The resulting two-pass procedure is:

1. run W-CIFSA with $\kappa = 0$;
2. collect valid pair-terminal marginal-stop states;
3. compute their finite κ_{req} values;
4. set κ^* to the 75th percentile;
5. rerun extraction using κ^* .

The rule is entirely structural and objective-based. No diffusion spread, cohesion metric, or downstream benchmark is used to select κ^* .

When W-CIFSA is compared with Strength-FSA or CI-FSA under the same (G, ℓ, λ) configuration, the W-CIFSA-derived κ^* is reused for those baselines rather than separately tuning a support parameter for each method. This avoids giving competing methods different degrees of regularization.

C.5 Calibration Sensitivity Analysis

We evaluated the calibration rule against the unregularized objective, a fixed $\kappa = 1$, and alternative quantile-based rules. The main empirical pattern was consistent across the benchmark networks. With $\kappa = 0$, extraction frequently terminated at pairs. Increasing κ produced progressively larger groups, confirming that the regularizer modifies group support smoothly rather than imposing a hard size constraint.

The median-based rule $Q_{0.50}$ often provided insufficient correction and left a substantial number of pair-terminal groups. In contrast, a fixed $\kappa = 1$ could over-correct, particularly when λ was large, causing substantially larger groups than were required to address the original pair bias. The pair-terminal $Q_{0.75}$ rule provided a more stable intermediate correction and was therefore adopted as the frozen calibration rule.

This behavior also explains why $\lambda = 0.75$ is treated as a stress configuration in the empirical analysis. When the objective places strong weight on internal

support, regularization and cohesion can interact to produce larger expansions. The primary operating range $\lambda \in \{0, 0.25, 0.50\}$ was substantially more stable.

The calibration should therefore be interpreted as a data-derived correction for small-set bias, not as a mechanism for enforcing a preferred group size. Its purpose is to reopen borderline pair-terminal states while preserving the greedy objective and the graph’s own local stopping structure.

D Algorithmic and Implementation Details

D.1 Complete W-CIFSA Algorithm

W-CIFSA consists of three stages: weighted collective-influence computation, structural calibration of the support parameter, and greedy connected group extraction. Algorithm 1 summarizes the complete two-pass procedure.

The first extraction pass is used only to estimate κ^* . The final reported groups are those obtained in the second pass. Seed ordering is based on WCI, while subsequent expansion is determined by the group-level candidate utility rather than by WCI alone.

D.2 Message-Passing Implementation

Explicit enumeration of all length- ℓ non-backtracking walks is avoided. We maintain one message for each directed edge. Initialization uses $M_{i \rightarrow j}^{(0)} = b_j$. For each propagation depth $r \geq 1$, we first compute the aggregate outgoing message at node j , $A_j^{(r)} = \sum_{x \in N(j)} p_{jx} M_{j \rightarrow x}^{(r-1)}$, and then remove the immediate backtracking contribution:

$$M_{i \rightarrow j}^{(r)} = A_j^{(r)} - p_{ji} M_{j \rightarrow i}^{(r-1)}.$$

After $\ell - 1$ iterations,

$$WCI_\ell(i) = b_i \sum_{j \in N(i)} p_{ij} M_{i \rightarrow j}^{(\ell-1)}.$$

An implementation should store directed edges explicitly or maintain paired indices for the two orientations of each undirected edge. This allows the reverse-edge message $M_{j \rightarrow i}$ to be retrieved in constant time. For numerical efficiency, only two message arrays are required: one for iteration $r - 1$ and one for iteration r , together with one node-sized array for $A_j^{(r)}$.

D.3 Lazy Heap-Based Group Expansion

For a current group S , each valid frontier node x is associated with attachment $a_x(S)$ and candidate utility $K_S(x) = (1 - \lambda)q_x + \frac{2\lambda}{s_{\max}} a_x(S)$. When a node y is added,

Algorithm 1 W-CIFSA**Require:** Weighted graph $G = (V, E)$, effective weights p_{ij} , radius ℓ , trade-off λ **Ensure:** Set of disjoint connected focal groups

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1: Compute  $s_i$  and  $b_i$  for all  $i \in V$ 
2: Compute  $WCI_\ell(i)$  by non-backtracking message passing
3: Normalize  $q_i = WCI_\ell(i) / \max_v WCI_\ell(v)$ 
4: Run greedy extraction with  $\kappa = 0$ 
5: Collect valid pair-terminal marginal-stop states  $\mathcal{T}_2$ 
6: if  $\mathcal{T}_2$  contains finite positive- $r_{\text{stop}}$  states then
7:   Compute  $\kappa_{\text{req}}(S) = 2(1/r_{\text{stop}}(S) - 1)$ 
8:   Set  $\kappa^* = Q_{0.75}(\{\kappa_{\text{req}}(S)\})$ 
9: else
10:   Set  $\kappa^* = 0$ 
11: end if
12: Reinitialize node assignments
13: Order seeds by decreasing  $WCI_\ell$ 
14: for each unassigned seed  $v$  do
15:   Initialize  $S \leftarrow \{v\}$ 
16:   Initialize its valid frontier and candidate utilities
17:   while the frontier is nonempty do
18:     Retrieve valid candidate  $x$  with maximum  $K_S(x)$ 
19:     if  $K_S(x) \leq F_{\lambda, \kappa^*}(S)$  then
20:       break
21:     end if
22:      $S \leftarrow S \cup \{x\}$ 
23:     Update only candidates adjacent to  $x$ 
24:     Add newly exposed valid frontier nodes
25:   end while
26:   if  $|S| \geq 2$  then
27:     Store  $S$  and mark its nodes assigned
28:   end if
29: end for
30: return discovered focal groups

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$$a_x(S \cup \{y\}) = a_x(S) + p_{xy},$$

and therefore

$$K_{S \cup \{y\}}(x) = K_S(x) + \frac{2\lambda}{s_{\max}} p_{xy}.$$

Only neighbors of y can change utility. W-CIFSA exploits this locality using a max-heap over frontier candidates.

Rather than performing an expensive in-place heap update, the implementation uses versioned lazy updates. Each node stores its current version number and latest candidate utility. Whenever its utility changes, the version is incremented and a new heap entry is inserted. When an entry is popped, it is accepted

only if its stored version equals the node’s current version and the node remains a valid frontier candidate. Older entries are discarded as stale.

A typical heap entry contains

$$(-K_S(x), x, \text{version}(x)),$$

when implemented with a standard min-heap. Ties should be resolved deterministically, for example by node identifier, to ensure reproducibility.

This strategy preserves the exact greedy ordering while avoiding global re-computation of all candidate utilities after each accepted node.

D.4 Numerical and Edge-Case Handling

Several implementation cases require explicit handling.

Zero-strength nodes. If $s_i = 0$, set $b_i = 0$. Such nodes receive zero WCI unless they enter through a separate preprocessing or feasibility rule.

Zero maximum WCI. If $\max_i WCI_\ell(i) = 0$, set $q_i = 0$ for all nodes rather than dividing by zero.

Zero $s_{\max}$. If the graph contains no positive effective edge weight, then $s_{\max} = 0$ and the weighted group objective is undefined in its usual form. Such a graph contains no meaningful weighted interaction structure and should be treated as a degenerate input case.

Pair-terminal calibration. Only two-node groups that stop because the best valid frontier candidate fails the marginal test are used for calibration. Empty-frontier and feasibility-induced stops are excluded. States with $r_{\text{stop}} = 0$ are logged but omitted from the finite κ_{req} distribution.

Empty calibration set. If no valid finite pair-terminal state exists, use $\kappa^* = 0$.

Floating-point comparison. The theoretical acceptance rule is $K_S(x) > F_{\lambda, \kappa}(S)$. In implementation, a small numerical tolerance may be used to avoid accepting changes caused only by floating-point roundoff. The same tolerance must be used consistently in extraction and terminal-state diagnostics.

Disconnected input graph. W-CIFSA may be applied independently across connected components. Because group growth is frontier-based, no focal group can span disconnected components.

Disjoint extraction. Once a group is finalized, assigned nodes are unavailable to subsequent groups. Candidate validity must therefore be rechecked when stale heap entries are popped.

Determinism. Randomness is not required by the core W-CIFSA algorithm. Fixed node-order tie breaking should be used for WCI ties, candidate-utility ties, and equal calibration values so repeated runs are reproducible.

D.5 Implementation Verification

We performed several independent checks before running the empirical evaluation.

Weighted branching. The implementation of b_i was checked against direct calculation on hand-constructed weighted graphs. Unit-weight graphs recover $b_i = k_i - 1$, and weighted leaves produce $b_i = 0$.

WCI reduction. For unit-weight graphs and $\ell = 1$, computed WCI values were verified to match classical CI_1 exactly.

Non-backtracking messages. For small graphs, message-passing WCI was compared with explicit enumeration of non-backtracking walks. The two computations agreed numerically. Each propagation depth processes every directed edge once, giving $O(\ell m)$ time and $O(m + n)$ memory.

Objective and marginal gain. For randomly generated weighted graphs and connected sets S , the closed-form marginal expression was compared with direct recomputation of $F_{\lambda, \kappa}(S \cup \{x\}) - F_{\lambda, \kappa}(S)$.

Heap-based extraction. The versioned lazy-heap implementation was compared against a brute-force greedy implementation that recomputed all valid frontier candidate utilities at every iteration. Across 200 randomly generated weighted graphs, both implementations produced the same accepted node sequence, terminal group, and objective value.

Classical CI baseline. The classical CI_ℓ implementation was independently checked to ensure that it uses shortest-distance spheres,

$$CI_\ell(i) = (k_i - 1) \sum_{j: d(i, j) = \ell} (k_j - 1),$$

rather than the unit-weight non-backtracking WCI formulation. This distinction is required for $\ell > 1$ on cyclic graphs.

Group matching and rank statistics. Experimental utilities were also checked independently. Group comparison uses size-aware matching rather than index-based matching, and rank correlations use tie-aware implementations.

Terminal local optimality. In the synthetic exact-optimality diagnostics, all 9,600 terminal groups were independently rechecked against every valid frontier addition. No improving one-node augmentation was found, confirming consistency between the production stopping rule and the independent diagnostic implementation.

These verification steps test implementation correctness rather than theoretical optimality. Exact global comparisons on F2F and synthetic graphs are reported separately in Appendix ??.

E Datasets and Preprocessing

E.1 Network Sources and Weight Semantics

The empirical evaluation uses three primary weighted networks: USAir97, Reddit, and MathOverflow. These networks differ in domain, scale, and the meaning of their edge weights, allowing W-CIFSA to be evaluated beyond a single type of weighted interaction.

USAir97. USAir97 is a weighted network of U.S. airports and air-transportation connections. Nodes represent airports and edges represent connections between airports; the supplied edge values encode connection intensity. We use USAir97 as the compact real-network benchmark because its size permits computationally expensive influence-maximization comparisons.

Reddit. The Reddit network is derived from the subreddit hyperlink dataset [3]. Nodes represent subreddits and edges are obtained from observed hyperlink interactions between communities. The original data are interaction-derived rather than a network with intrinsic physical edge weights. We aggregate repeated interactions between node pairs to obtain a weighted undirected graph for the present analysis.

MathOverflow. MathOverflow is derived from the temporal Stack Exchange interaction data [5]. Nodes represent users and temporal records encode interactions among them. Repeated interactions are aggregated to form the weighted undirected network used in our experiments. As with Reddit, the edge weights therefore represent interaction frequency rather than a directly measured transmission probability.

Small F2F resistance networks are used only for exact connected-subset optimization. Because these graphs are sufficiently small for exhaustive enumeration, they provide a real weighted-network check of greedy solution quality rather than a primary diffusion benchmark. Synthetic weighted networks are used separately to test exact optimality across controlled topological and weight regimes; their generation procedure is described in Appendix ??.

F Experimental Protocols

F.1 Baselines and Group Matching

We compare W-CIFSA with structural, ranking-based, community-based, and influence-maximization baselines. Structural baselines include Strength-FSA and CI-FSA, which use the same connected group-extraction procedure as W-CIFSA but replace WCI with weighted strength and classical collective influence, respectively. Ranking baselines include top- k weighted strength, WCI, PageRank, and eigenvector centrality. We also include the community-aware influential-node method of Rajeh and Cherifi [6], together with weighted Leiden [?] and Infomap [7]. Random connected groups provide a structural reference.

For direct influence-maximization comparison, CELF [4] is used on USAir97, while scalable reverse-influence sampling [1] is used on Reddit and MathOverflow. All group-level comparisons are size matched: for each W-CIFSA group of size k , the competing method is evaluated using exactly k nodes. For methods that naturally return multiple groups, matching is performed by group size rather than output order. Community baselines are restricted to feasible connected k -node subsets when an exact-size comparison is required.

F.2 Diffusion and Influence-Maximization Evaluation

Diffusion performance is evaluated with the independent-cascade model [2]. For each seed set S , expected spread $\sigma(S)$ is estimated using the same final Monte Carlo protocol across all methods. The graph, propagation probabilities, random-seed policy, and number of simulation trials are held fixed within each comparison.

CELF is used where repeated Monte Carlo marginal-gain estimation is tractable. On the larger Reddit and MathOverflow networks, reverse-influence sampling is used to obtain scalable influence-maximizing reference sets. W-CIFSA does not use diffusion outcomes during group discovery or support calibration; diffusion is used only for post hoc evaluation.

For each matched seed size, we report both absolute or normalized spread and spread retention relative to the corresponding influence-maximization solution,

$$RR_{\text{IM}} = \frac{\sigma(S_{\text{WCI}})}{\sigma(S_{\text{IM}})}.$$

Values near one indicate that the connected W-CIFSA group retains most of the spread achieved by the diffusion-optimized seed set.

F.3 Lower-Saturation and Robustness Conditions

Because high propagation probabilities can produce saturated cascades, we also evaluate weaker diffusion regimes in which seed selection has greater influence on cascade size. W-CIFSA groups are frozen from the primary structural selection. For each lower-propagation condition, the influence-maximization baseline

is reoptimized under that condition and the resulting seed sets are evaluated using the same Monte Carlo protocol.

We record the activated fraction and spread dispersion for each regime to verify that the condition is genuinely discriminative. The lowest-propagation setting targets a mean edge activation probability near 0.01; additional intermediate conditions provide a gradual transition from the primary model to this low-saturation regime.

Weight-robustness experiments alter only the weights used during W-CIFSA selection. Uniform weights remove heterogeneity, shuffled weights preserve the marginal weight distribution but destroy structural placement, multiplicative noise simulates measurement error, and progressive homogenization moves weights toward their network mean. Resulting groups are evaluated on the unchanged reference graph so that selection robustness is not conflated with a changed evaluation network.

F.4 Metrics, Runtime, and Hardware

For a seed set S , normalized spread is

$$NS(S) = \frac{\sigma(S) - |S|}{|V| - |S|}.$$

Let $A(S)$ denote total internal edge weight and $W_{\text{out}}(S)$ the total edge weight crossing from S to $V \setminus S$. Internal structure is summarized by

$$R(S) = \frac{2A(S)}{2A(S) + W_{\text{out}}(S)}, \quad D_w(S) = \frac{2A(S)}{|S|(|S| - 1)},$$

and weighted conductance

$$\phi(S) = \frac{W_{\text{out}}(S)}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}.$$

Connectivity and number of connected components are reported separately where the baseline does not impose connectivity. We do not combine diffusion and internal structure into a single composite score; comparisons instead use matched metrics and Pareto analysis.

For scalability experiments, we measure WCI computation, first-pass extraction, support calibration, second-pass extraction, total runtime, and peak resident memory. Timing experiments use a fixed CPU/thread configuration, with repeated runs summarized by the median where feasible. Hardware, software versions, random seeds, Monte Carlo counts, and network-specific runtime settings are reported in Table 1.

G Supplementary Influence-Maximization Results

All comparisons use matched seed-set cardinality. Connectivity is reported separately from retention, density, and conductance. Community rows marked “or-

acle subset” use an exact-size community candidate post-selected by simulated spread.

Network	Diffusion	Method	NS	R	n
Reddit	mean p=0.01	CI-FSA	0.0163	0.0184	30
Reddit	mean p=0.01	Infomap (oracle subset)	0.0165	0.0317	24
Reddit	mean p=0.01	Leiden (oracle subset)	0.0166	0.0181	24
Reddit	mean p=0.01	Random connected	0.0108	0.0444	600
Reddit	mean p=0.01	RC-PageRank	0.0166	0.00691	24
Reddit	mean p=0.01	RC-Strength	0.0164	0.00886	24
Reddit	mean p=0.01	RIS	0.0165	0	30
Reddit	mean p=0.01	Strength-FSA	0.0164	0.0182	30
Reddit	mean p=0.01	Eigenvector	0.0163	0.0134	24
Reddit	mean p=0.01	PageRank	0.0165	0.00911	24
Reddit	mean p=0.01	Top-k strength	0.0163	0.0113	24
Reddit	mean p=0.01	Top-k WCI	0.0163	0.0122	24
Reddit	mean p=0.01	W-CIFSA	0.0163	0.0144	30
Reddit	primary	CI-FSA	0.206	0.0241	40
Reddit	primary	RIS	0.207	0	40
Reddit	primary	Strength-FSA	0.206	0.0217	40
Reddit	primary	W-CIFSA	0.206	0.0304	40
MathOverflow	mean p=0.01	CI-FSA	0.0526	0.00729	30
MathOverflow	mean p=0.01	Random connected	0.0447	0.0271	450
MathOverflow	mean p=0.01	RIS	0.0527	0	30
MathOverflow	mean p=0.01	Strength-FSA	0.0526	0.00641	30
MathOverflow	mean p=0.01	Eigenvector	0.0526	0.00525	18
MathOverflow	mean p=0.01	PageRank	0.0527	0.00456	18
MathOverflow	mean p=0.01	Top-k strength	0.0527	0.00424	18
MathOverflow	mean p=0.01	Top-k WCI	0.0527	0.00503	18
MathOverflow	mean p=0.01	W-CIFSA	0.0527	0.00515	30
MathOverflow	primary	CI-FSA	0.335	0.00795	40
MathOverflow	primary	RIS	0.335	0	40
MathOverflow	primary	Strength-FSA	0.335	0.00641	40
MathOverflow	primary	W-CIFSA	0.335	0.00548	40
USAir97	mean p=0.01	CI-FSA	0.0116	0.0352	30
USAir97	mean p=0.01	Connected-IM	0.0158	0.0132	9
USAir97	mean p=0.01	Infomap (oracle subset)	0.0163	0.0134	9
USAir97	mean p=0.01	Leiden (oracle subset)	0.0163	0.0134	9
USAir97	mean p=0.01	Random connected	0.00547	0.0461	225
USAir97	mean p=0.01	RC-PageRank	0.0156	0.0252	9
USAir97	mean p=0.01	RC-Strength	0.0155	0.0181	9
USAir97	mean p=0.01	RIS	0.0138	0.0169	30
USAir97	mean p=0.01	Strength-FSA	0.015	0.0405	30
USAir97	mean p=0.01	Eigenvector	0.0146	0.0142	9
USAir97	mean p=0.01	PageRank	0.016	0.0123	9

Network	Diffusion	Method	NS	R	n
USAir97	mean $p=0.01$	Top-k strength	0.0163	0.0134	9
USAir97	mean $p=0.01$	Top-k WCI	0.0161	0.0167	9
USAir97	mean $p=0.01$	W-CIFSA	0.0135	0.0231	30
USAir97	primary	CELF	0.337	0.00124	34
USAir97	primary	Connected-IM	0.331	0.0643	34
USAir97	primary	W-CIFSA	0.32	0.104	40

H Lower-Saturation Diffusion Robustness

Reddit retained a mean 98.8% (minimum 96.7%) of reoptimized RIS spread at $\bar{p} \simeq 0.01$. MathOverflow retained a mean 99.9% (minimum 99.7%) of re-optimized RIS spread at $\bar{p} \simeq 0.01$. These statements apply only to the tested lower-propagation conditions.

Network	Condition	ℓ	mean RR_{RIS}	min RR_{RIS}	n
Reddit	saturating_mean_0.010	1	0.993	0.987	15
Reddit	saturating_mean_0.010	2	0.982	0.967	15
Reddit	saturating_mean_0.025	1	0.997	0.995	15
Reddit	saturating_mean_0.025	2	0.992	0.985	15
Reddit	saturating_mean_0.050	1	0.997	0.996	15
Reddit	saturating_mean_0.050	2	0.996	0.994	15
Reddit	scaled_0.25	1	0.999	0.997	15
Reddit	scaled_0.25	2	0.996	0.991	15
Reddit	scaled_0.50	1	0.999	0.997	15
Reddit	scaled_0.50	2	0.997	0.995	15
Reddit	scaled_0.75	1	0.999	0.998	15
Reddit	scaled_0.75	2	0.999	0.997	15
Reddit	scaled_1.00	1	0.999	0.998	15
Reddit	scaled_1.00	2	0.998	0.997	15
MathOverflow	saturating_mean_0.010	1	1	0.998	15
MathOverflow	saturating_mean_0.010	2	0.998	0.997	15
MathOverflow	saturating_mean_0.025	1	0.999	0.998	15
MathOverflow	saturating_mean_0.025	2	0.999	0.999	15
MathOverflow	saturating_mean_0.050	1	0.999	0.999	15
MathOverflow	saturating_mean_0.050	2	0.999	0.999	15
MathOverflow	scaled_0.25	1	0.999	0.999	15
MathOverflow	scaled_0.25	2	0.999	0.999	15
MathOverflow	scaled_0.50	1	0.999	0.999	15
MathOverflow	scaled_0.50	2	1	0.999	15
MathOverflow	scaled_0.75	1	1	0.999	15
MathOverflow	scaled_0.75	2	1	1	15
MathOverflow	scaled_1.00	1	1	0.999	15

Network	Condition	ℓ	mean RR_{RIS}	min RR_{RIS}	n
MathOverflow	scaled_1.00	2		1	1 15

I Exact Optimality Analysis

Across 9600 synthetic configurations, mean OR was 0.986 (median 1, minimum 0.62); across 992 F2F configurations, mean OR was 1. Best-seed diagnostics increased mean OR to 0.997; 95.0% exactly matched the optimum, and 17.5% improved over the production seed. The enumeration study is diagnostic and does not establish a global approximation guarantee.

Set	mean OR	median OR	min OR	exact rate	n
F2F	1	1	0.997	0.976	992
synthetic	0.986	1	0.62	0.788	9600

Family	Add	Delete	Swap	n
ba	–	–	–	1600
bridge	–	–	–	1600
core_periphery	–	–	–	1600
er	–	–	–	1600
hub_bridge	–	–	–	1600
sbm	–	–	–	1600

J Weight Ablation and Perturbation Robustness

Perturbed-weight groups were evaluated on the reference graph. noise_0.2 gave mean $\rho = 0.998$ and mean spread retention 1.03. noise_0.4 gave mean $\rho = 0.996$ and mean spread retention 0.998. reference gave mean $\rho = 1$ and mean spread retention 1. shuffle gave mean $\rho = 0.976$ and mean spread retention 1. uniform gave mean $\rho = 0.985$ and mean spread retention 1.01. Ranking, membership, diffusion, cohesion, and calibration outputs are retained separately; no composite score is used.

Network	Condition	ρ	τ	J spread retention
Reddit	noise_0.05	0.988	0.977	0.883
Reddit	noise_0.1	0.988	0.973	0.947
Reddit	reference	1	1	1
Reddit	shuffle	0.994	0.948	0.544
Reddit	uniform	0.997	0.964	0.557
MathOverflow	noise_0.05	0.999	0.989	0.965
MathOverflow	noise_0.1	0.999	0.981	0.925
MathOverflow	noise_0.2	0.997	0.964	0.98

Network	Condition	ρ	τ	J spread	retention
MathOverflow	reference	1	1	1	1
MathOverflow	shuffle	0.985	0.909	0.423	1
MathOverflow	uniform	0.994	0.942	0.399	1
USAir97	homogenize_0.0	1	1	1	1
USAir97	homogenize_0.25	0.997	0.961	0.607	1.09
USAir97	homogenize_0.5	0.99	0.926	0.415	1.08
USAir97	homogenize_0.75	0.98	0.895	0.351	1
USAir97	homogenize_1.0	0.964	0.859	0.298	0.994
USAir97	noise_0.05	0.999	0.992	0.807	1.04
USAir97	noise_0.1	0.999	0.986	0.712	1.05
USAir97	noise_0.2	0.998	0.975	0.571	1.03
USAir97	noise_0.4	0.996	0.954	0.394	0.998
USAir97	reference	1	1	1	1
USAir97	shuffle	0.948	0.821	0.259	1.01
USAir97	uniform	0.964	0.859	0.305	1.03

K Supplementary Scalability Analysis

Reddit had fitted slope $b = 0.86$ (95% CI 0.745–0.974, $R^2 = 0.927$). MathOverflow had fitted slope $b = 0.905$ (95% CI 0.797–1.01, $R^2 = 0.941$). The log–log fits summarize empirical runtime behavior and do not prove an asymptotic bound. Normalized WCI and extraction timings are provided in machine-readable form.

Network	b 95% CI	R^2	n
Reddit	0.86 [0.745, 0.974]	0.927	19
MathOverflow	0.905 [0.797, 1.01]	0.941	19

L Additional Sensitivity Analyses

Radius, trade-off, support-regularization, and stress-setting summaries reuse completed experiments without retuning the production method.

Network	Diffusion	ℓ	λ	NS	R
Reddit	mean p=0.01	1	0.0	0.0163	0.0105
Reddit	mean p=0.01	1	0.25	0.0163	0.0105
Reddit	mean p=0.01	1	0.5	0.0163	0.0114
Reddit	mean p=0.01	2	0.0	0.0163	0.0165
Reddit	mean p=0.01	2	0.25	0.0162	0.0184
Reddit	mean p=0.01	2	0.5	0.0162	0.0189
Reddit	primary	1	0.0	0.207	0.0105
Reddit	primary	1	0.25	0.206	0.0105
Reddit	primary	1	0.5	0.207	0.0114

Network	Diffusion	ℓ	λ	NS	R
Reddit	primary	1	0.75	0.206	0.0799
Reddit	primary	2	0.0	0.207	0.0165
Reddit	primary	2	0.25	0.207	0.0184
Reddit	primary	2	0.5	0.206	0.0189
Reddit	primary	2	0.75	0.206	0.0774
MathOverflow	mean p=0.01	1	0.0	0.0526	0.00558
MathOverflow	mean p=0.01	1	0.25	0.0526	0.00558
MathOverflow	mean p=0.01	1	0.5	0.0526	0.00558
MathOverflow	mean p=0.01	2	0.0	0.0527	0.00472
MathOverflow	mean p=0.01	2	0.25	0.0527	0.00472
MathOverflow	mean p=0.01	2	0.5	0.0527	0.00472
MathOverflow	primary	1	0.0	0.335	0.00558
MathOverflow	primary	1	0.25	0.335	0.00558
MathOverflow	primary	1	0.5	0.335	0.00558
MathOverflow	primary	1	0.75	0.335	0.00659
MathOverflow	primary	2	0.0	0.335	0.00472
MathOverflow	primary	2	0.25	0.335	0.00472
MathOverflow	primary	2	0.5	0.335	0.00472
MathOverflow	primary	2	0.75	0.335	0.00635
USAir97	mean p=0.01	1	0.0	0.0124	0.0138
USAir97	mean p=0.01	1	0.25	0.0131	0.016
USAir97	mean p=0.01	1	0.5	0.0179	0.0545
USAir97	mean p=0.01	2	0.0	0.0124	0.0155
USAir97	mean p=0.01	2	0.25	0.0124	0.0155
USAir97	mean p=0.01	2	0.5	0.0128	0.0235
USAir97	primary	1	0.0	0.326	0.0461
USAir97	primary	1	0.25	0.325	0.0534
USAir97	primary	1	0.5	0.32	0.113
USAir97	primary	1	0.75	0.309	0.194
USAir97	primary	2	0.0	0.325	0.0597
USAir97	primary	2	0.25	0.325	0.0621
USAir97	primary	2	0.5	0.32	0.116
USAir97	primary	2	0.75	0.312	0.186

M Supplementary Real-World Case Study Material

The frozen Indonesia 2025 interaction graph contains 12,112 nodes and 15,982 undirected edges derived from 20,462 retained tweets. Sample text supplies interaction context only and does not establish coordination, affinity, endorsement, motivation, shared intent, or causal influence.

Group	$ S $	$\sum$	WCI	R	D_w	ϕ Leiden	C
G0002	6	0.606	0.0346	0.17	0.965		5
G0135	17	0.00142	0.343	0.0181	0.657		8
G0160	14	0.00079	0.731	0.023	0.269		1
G0194	55	0.00107	0.864	0.0056	0.136		4
G0592	41	7.73e-05	0.943	0.00773	0.0572		4

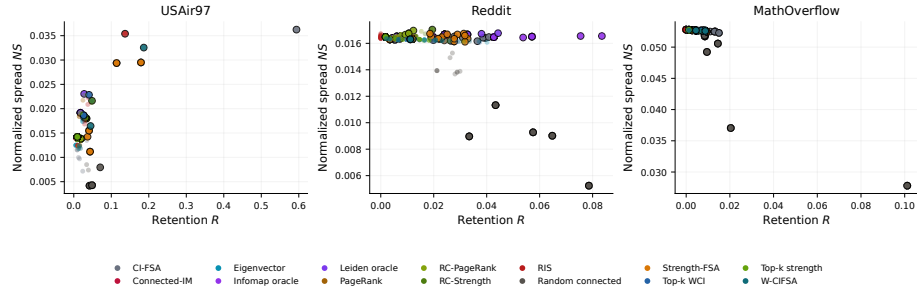


Fig. 1. Influence-cohesion Pareto configurations under the matched lower-saturation condition. Black-edged points are nondominated within network, radius, trade-off, budget, and diffusion condition.

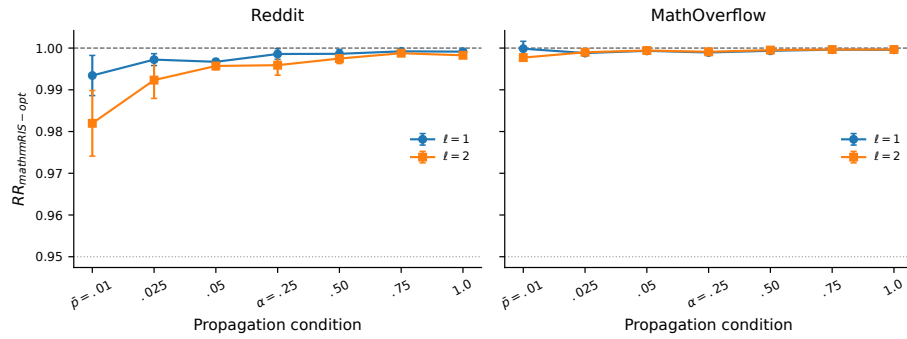


Fig. 2. Spread retention relative to condition-specific reoptimized RIS. Error bars show standard deviation across matched configurations.

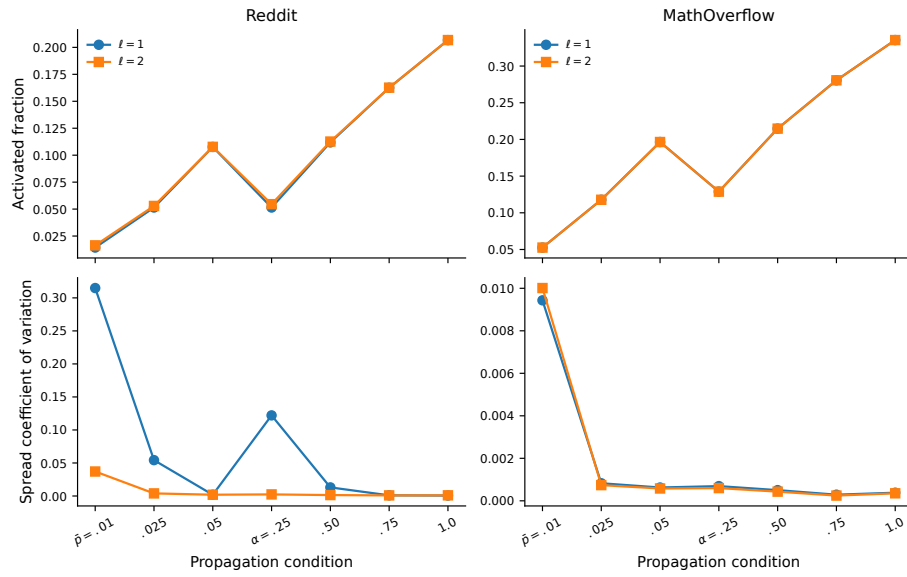


Fig. 3. Activation and spread-variation diagnostics across propagation conditions.

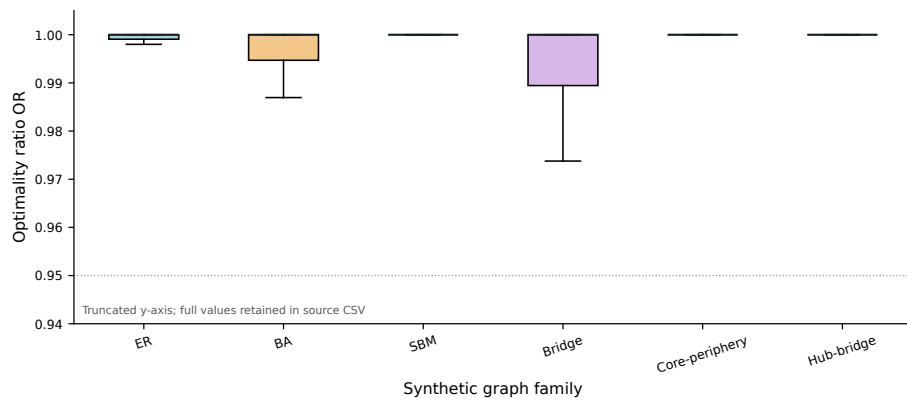


Fig. 4. Synthetic exact-enumeration optimality ratios by topology.

Table 1. Computational environment and network-specific runtime configuration. All reported influence spreads use common Monte Carlo streams within a matched comparison. The timeout applies to a complete CELF configuration, not to each seed budget.

Computational environment					
CPU	2 × Intel Xeon Gold 6248 (2.50 GHz); 40 physical cores, 80 threads				
Memory	1 TiB RAM (1007.1 GiB reported)				
Operating system	Ubuntu 22.04.5 LTS; Linux 5.15.0-181-generic; x86-64				
Software	Python 3.13.5; W-CIFSA 0.1.0; NumPy 2.2.6				
Community software	python-igraph 1.0.0; leidenalg 0.12.0; Infomap 2.15.1				
Plotting	Matplotlib 3.11.1				
Acceleration	CPU only; no GPU used				
Scalability timing	One thread; OMP/OpenBLAS/MKL/NumExpr threads fixed to 1				
Network	n	m	IM optimizer	MC/RR budget	Randomness Execution limit
USAir97	332	2,126	Connected-IM	100 MC per Base seed 20260825; common final final MC stream	7,200 s per CELF or Connected-IM configuration
Reddit	34,671	123,570	Heterogeneous RIS	10,000 sets; 1,000 final MC	RR RIS seed = CELF omit- ted after one representa- tive 7,200 s timeout $100\ell + 100\lambda +$ i
MathOverflow	24,668	187,939	Heterogeneous RIS	10,000 sets; 1,000 final MC	RR Same derived-seed rules as Reddit Six parallel community workers; CELF omit- ted after one representa- tive timeout
Shared model and evaluation settings					
W-CIFSA grid	$\ell \in \{1, 2\}$, $\lambda \in \{0, .25, .50, .75\}$, at most five groups; pair-terminal Q_{75} support calibration.				
Weights	USAir97 uses published weighted edges. Reddit and MathOverflow weights are aggregated interaction counts. W-CIFSA uses log-normalized weights.				
Primary diffusion	$p_{uv} = \min\{1, [\log(1 + w_{uv}) / \max_e \log(1 + w_e)]c\}$, with c chosen for mean edge probability 0.10.				
Community baselines	Weighted Leiden and Infomap use seed 20260825. Random-connected groups use seed 20260825+ k , 30 samples per matched size, and 1,000 MC simulations under $p_{uv} = 1 - \exp(-\beta w_{uv})$ calibrated to mean 0.01.				
Scalability study	Seed 20260831; three graph samples and three timing repeats per fraction; one warm-up; 7,200 s per sampled-graph configuration.				

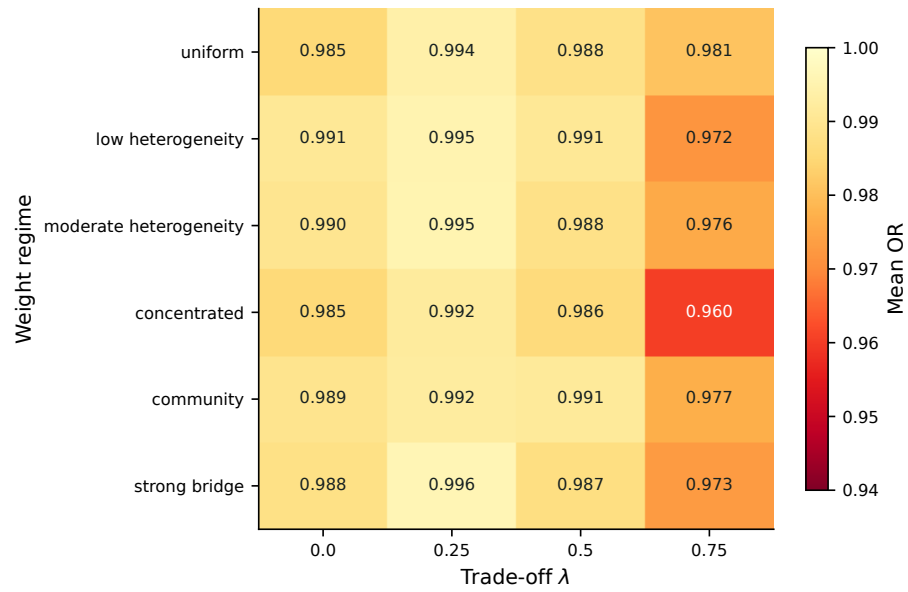


Fig. 5. Mean optimality ratio by trade-off and weight regime.

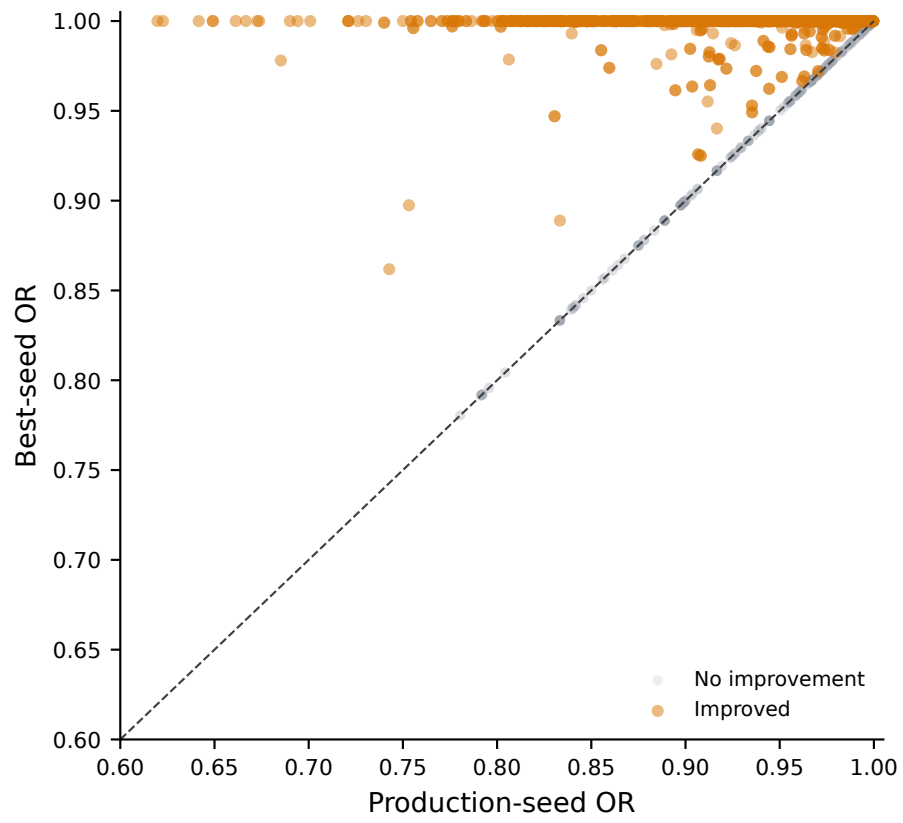


Fig. 6. Production-seed versus best-seed optimality ratios.

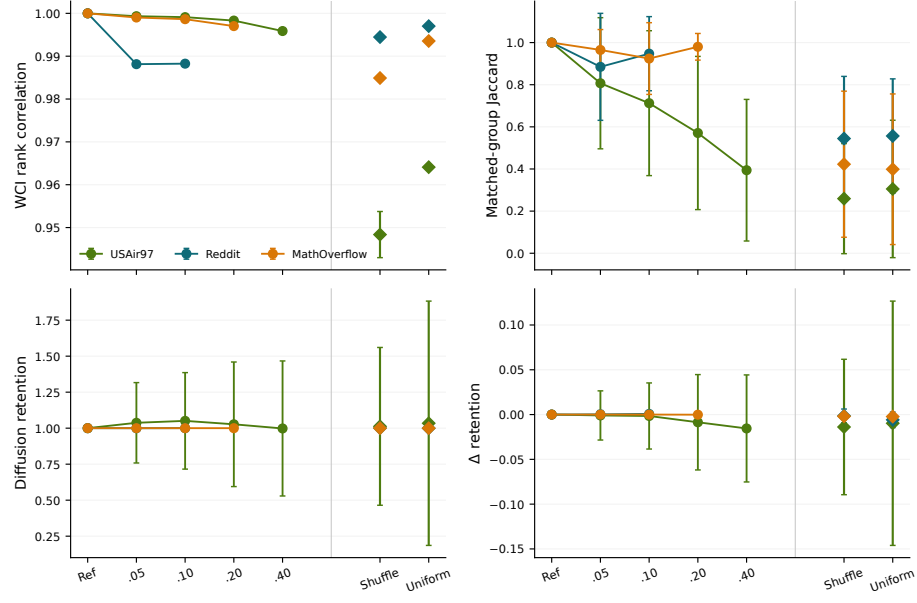


Fig. 7. Ranking, membership, diffusion, and cohesion stability under weight perturbation.

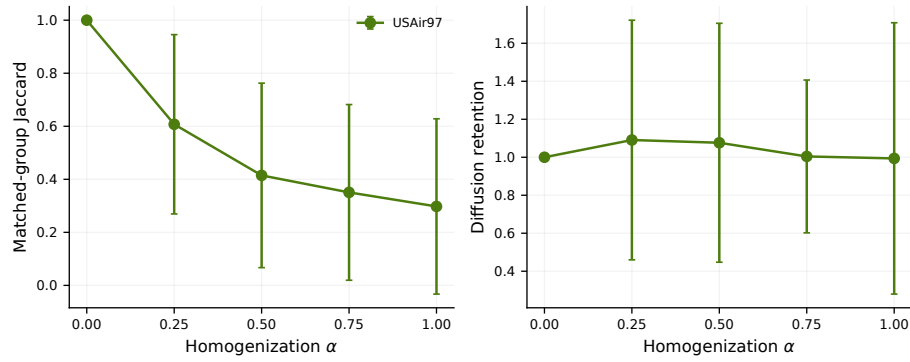


Fig. 8. Progressive weight homogenization; only completed datasets are shown.

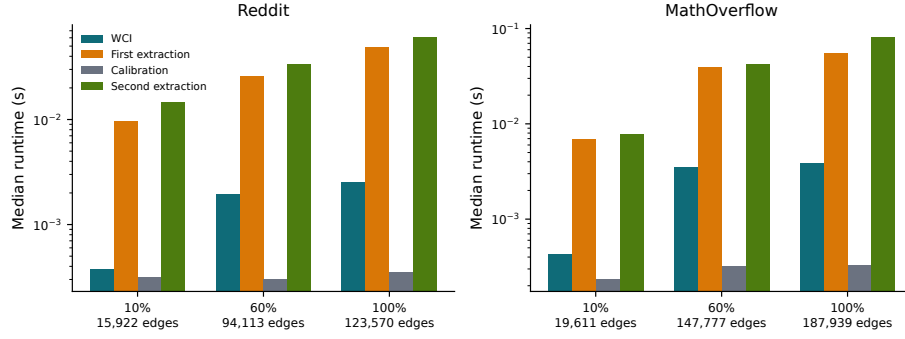


Fig. 9. Median runtime decomposition at representative graph sizes.

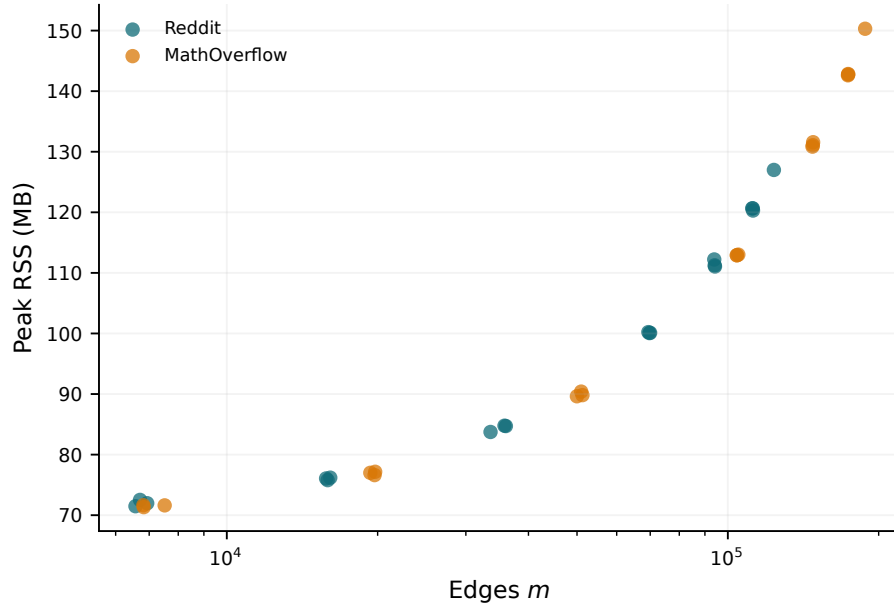


Fig. 10. Peak resident memory against edge count for independently sampled sub-graphs.

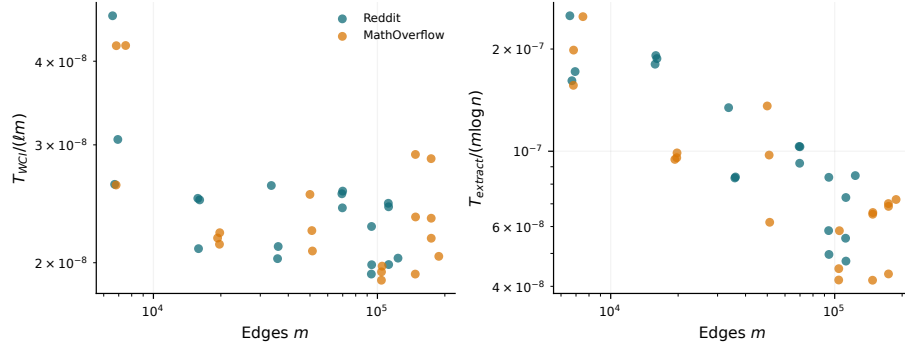


Fig. 11. Normalized WCI and extraction timing diagnostics.

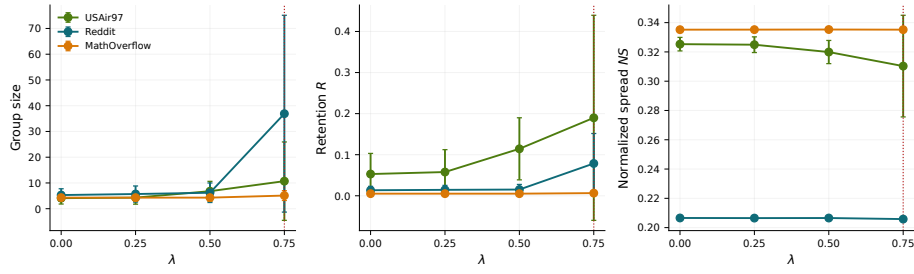


Fig. 12. Effect of the influence–interaction trade-off; the dotted line marks the stress setting.

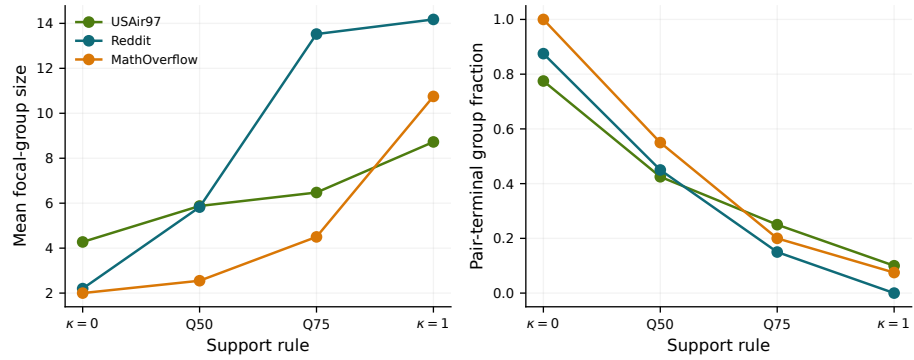


Fig. 13. Support-rule effects on focal-group size and pair frequency.

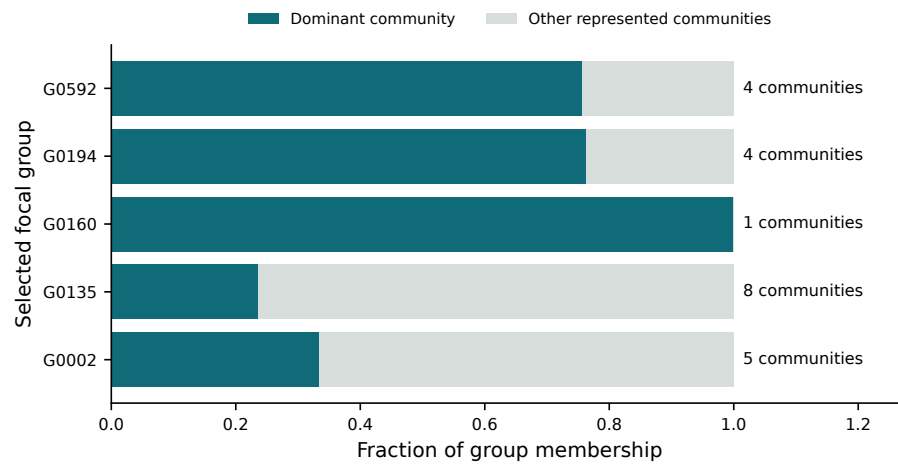


Fig. 14. Weighted Leiden composition of the selected case-study focal groups.

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